Species Narratives

Species Narratives summarize the vulnerability results for each species, and provide additional context on known climate change impacts on the species and known life history characteristics. PDF versions of the narratives will be available for download. All figures can be enlarged by clicking on the figure.

Each narrative contains the following information:

Summary

A summary section providing the final average Vulnerability, Exposure, and Sensitivity scores. If scores were calculated across species stocks and scores differed across stocks, a range of scores will be provided. Data quality scores associated with both Exposure and Sensitivity scores will also be provided.

This section also contains a summary table.

Changes from Previous Assessments

This section contains a table highlighting the scores from this assessment, and compares them to previous assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Stocks

If a species was scored within different stock groups, a map of the different stocks is provided here and referenced throughout the narrative.

Score Descriptions

These sections go into the scores in-depth, describing average scores, spatial patterns, and drivers. Each section has a map of scores. Climate Exposure has two sets of maps - one illustrating total exposure and the second illustrating factor exposures.

Climate Exposure also contains two subsections - one that describes monthly variability in both total and factor exposures, and a second that focuses on only factors that were important in the species distribution model.

Total scores were calculated spatially using the same logic rule as Hare et al. (2016):

Overall Sensitivity or Exposure Score	Numeric Score	Logic Rule	
Very High	4	3 or more attributes or factors	3.5
High	3	2 or more attributes or factors	3.0
Moderate	2	2 or more attributes or factors	2.5
Low	1	All other scores	

Overall vulnerability and exposure scores were calculated by averaging results across the target species' range and within defined stocks if available. Average overall scores were defined as the following:

Average Vulnerability or Exposure Score	Average Exposure Numeric Score	Average Vulnerability Numeric Score
Very High	3.5	10.5
High	< 3.5	< 10.5
Moderate	< 2.5	< 7.0
Low	< 1.5	< 3.5

Data Quality Metrics

This section describes the Life History Data Quality and Model Confidence Scores. Both were provided by the experts and range from 0 - 3. Following the original analysis, a score above 2 was deemed "acceptable". The rubrics for these scores are provided below

Life History Data Quality Rubric

Data Quality Score	Description
3	**Adequate Data.** The score is based on data which have been observed, modeled or empirically measured for the species in question and comes a from reputable source.
2	**Limited Data.** The score is based on data which has a higher degree of uncertainty. The data used to score the attribute may be based on related or similar species, come from outside the study area, or the reliability of the source may be limited.
1	**Expert Judgment.** The attribute score reflects the expert judgment of the reviewer and is based on their general knowledge of the species, or other related species, and their relative role in the ecosystem.
0	**No Data.** No information to base an attribute score on. Very little is known about the species or related species and there is no basis for forming an expert opinion (please use judiciously).

This is the same rubric used in Hare et al. (2016).

Model Confidence Rubric

Model Confidence Score	Description
3	**High Confidence.** The model highly accurately reflects the species' realized niche. It captures fine-scale nuances, matches expert field observations, and handles environmental transitions realistically.
2	**Moderate Confidence.** The model aligns well with known distributions and environmental drivers. Most key habitats are represented, though there may be minor inaccuracies at the range edges or in data-poor regions.
1	**Low Confidence.** The model captures broad trends but has significant "noise." Environmental drivers likely don't reflect actual species needs.

Model Confidence Score	Description
0	**No Confidence.** The model fails to represent known species range. It predicts species in unsuitable areas or misses well-known core areas entirely.

This rubric was developed to mimic the Life History Data Quality rubric.

The Area under the Receiver-Operator Curve (AUC) is also provided for the ensemble model with the Model Confidence Scores. This statistic quantifies how well models classify binary data. The closer the value is to one, the better the model did at predicting observed presences and absences. This metric was not provided to the experts while scoring Model Confidence.

Additional Metrics

This section describes the Distribution Shifts and Directional Effect results. Directional Effect is consistent with the previous analysis - experts were asked to score how they left the species would be impacted (postive/negative/neutral) by climate change.

Distribution Shifts were previously assessed using a subset of Sensitivity Attributes. In this analysis, they are quantified with the ensemble species distribution models when possible. The provided map illustrates average species ranges using 5 year averages from the first and last 5 years of the hindcast data and the last 5 years of the most recent decadal forecast. The provided timeseries illustrate how core and total (quantified by 50 and 95% Kernel Density Estimator (KDE) respectively) habitat areas, latitude range, and longitude ranges change throughout the MOM6 hindcast (black points) and decadal forecast (yellow points) datasets.

In the online version of the narratives, an animated gif is also provided to illustrate how species distributions and the edges of these distributions shift through time in both the hindcast and forecasts.

Climate Effects & Life History Synopsis

The final two sections of the narratives include summaries of known climate impacts on a species and a summary of their life histories.

References

Hare, Jonathan A, Wendy E Morrison, Mark W Nelson, Megan M Stachura, Eric J Teeters, Roger B Griffis, Michael A Alexander, et al. 2016. "A Vulnerability Assessment of Fish and Invertebrates to Climate Change on the Northeast US Continental Shelf." *PloS One* 11 (2): e0146756.